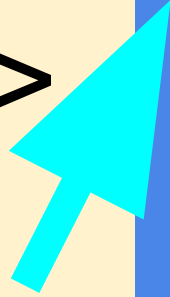


ONE PERSON IN GROUP-> CLICK USE TEMPLATE



WHEN THE COPIED TEMPLATE OF THESE SLIDES OPENS IN A NEW TAB:

- ADD YOUR Group NAME TO THE TITLE
- SHARE WITH GROUP MEMBERS (USE SHARE BUTTON)
- COPY THE TITLE AND LINK TO YOUR RECORDS (Digital folder/table of contents)
- Remember it is advisable to make a PERSONAL copy and link it when DONE

Solubility Curves

ESSENTIAL QUESTIONS

Remember, you should be able to thoroughly answer these AFTER learning through following resources.

- What are the different ways we can describe solutions based on their saturation?
- What information does a Solubility graph provide?
- What is the difference between a gas and a solid in aqueous solution as temperature rises?

Instructions

Follow through the **Engage to Apply** resources in order .

- Once complete, choose from Extend resources as interested.
- Record your learning in **these slides- Remember to keep track of where to find these slides for studying!**
- Complete the **Essential Questions Response in the Apply Section** .
- Try the Extend section and optional Apply practice as desired to deepen your understanding

Hyperdoc Elements

Click for quick access to:

- [Engage](#)
- [Explore](#)
- [Apply](#)
- [Extend](#)

Terminology - As you encounter terms new to you, define in your own words here, include examples (make them up!)



Part 1: ENGAGE

(5 minutes)

Engage

Watch the video on the right about **cold ice** with your group and discuss the below:

Make three (3) observations and one (1) inference based on your observations about what the video shows.





DISCUSS:

Is there a limit to the amount of solute that will dissolve in a certain amount of solvent?

Part 2: EXPLORE

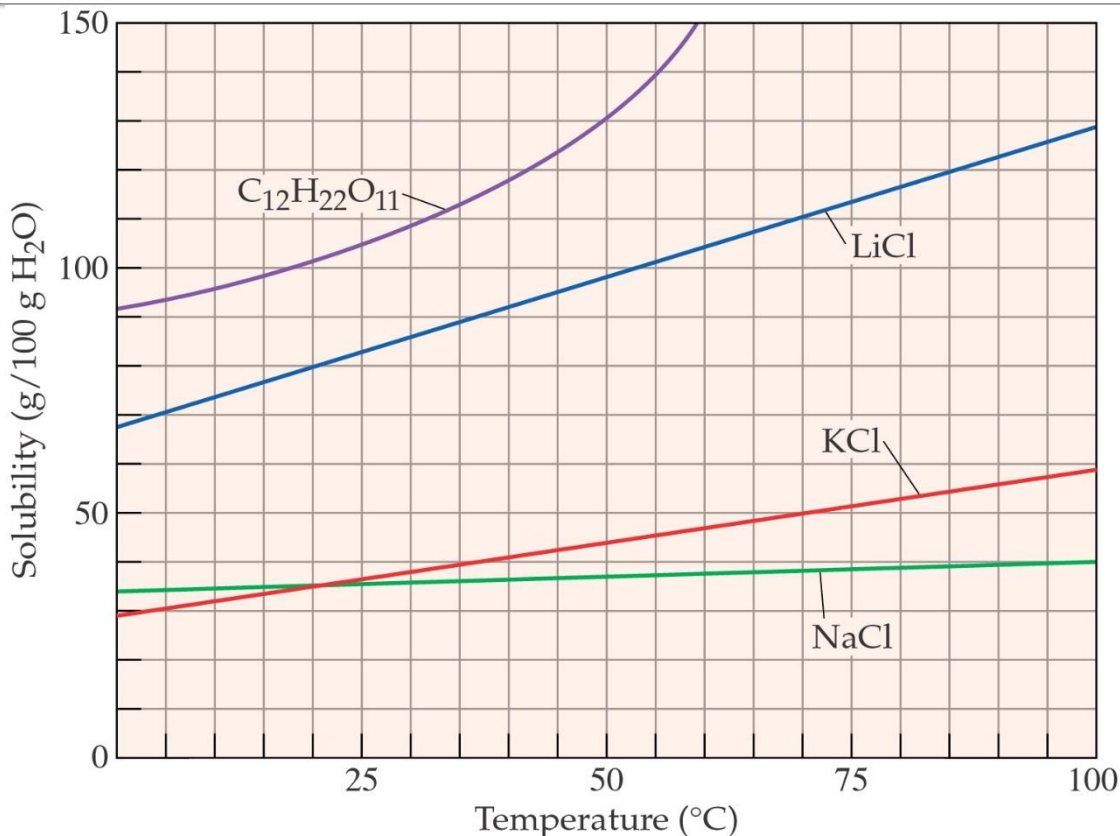
Explore: Solubility and temperature

- The *solubility* of a compound is the *maximum* amount of solute that can be dissolved in a solvent for a given set of conditions (ie temperature, pressure, etc)
 - It is often expressed in terms of 100g of solvent (note the units!)

Explore: Describing Solubility

*In general, a compound becomes more soluble as the temperature *increases*.*

Come up with a few examples of solutions you heat to increase solubility (think drinks).



Explore: Types of Solutions

Unsaturated Solution: As a group come up with a definition then move this box to compare

Saturated Solution: contains maximum amount of solute that can be dissolved in a given amount of solvent at a certain temperature.

Supersaturated Solution: As a group come up with a definition then move this box to compare

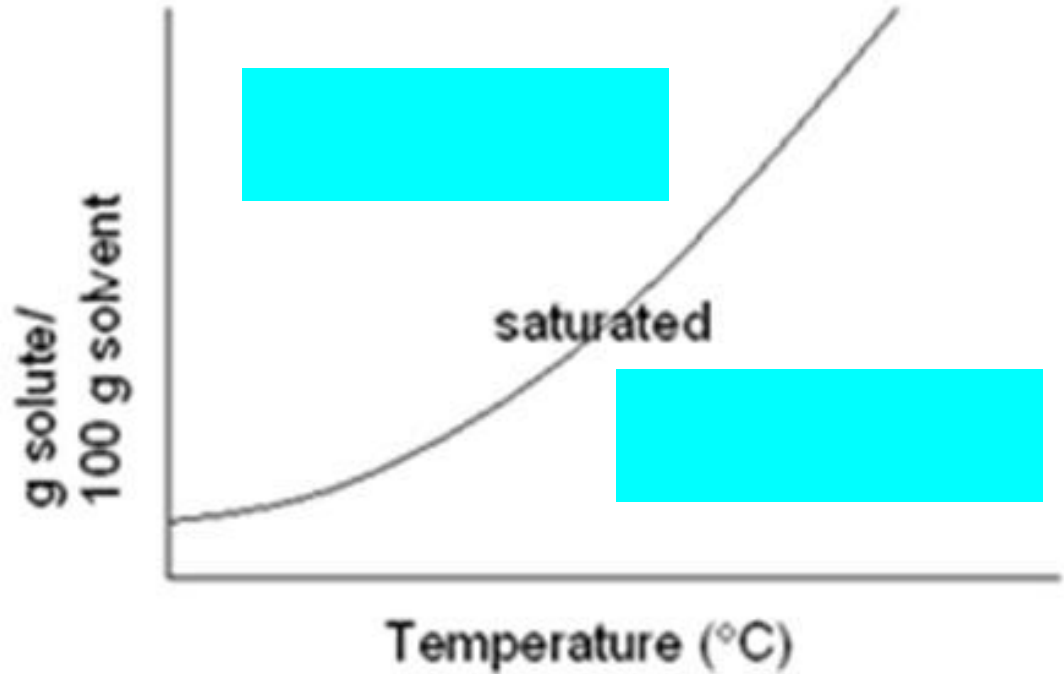


Optional: watch the video to enhance understanding

Explore: Describing Solubility

- Point on the line is **saturated**

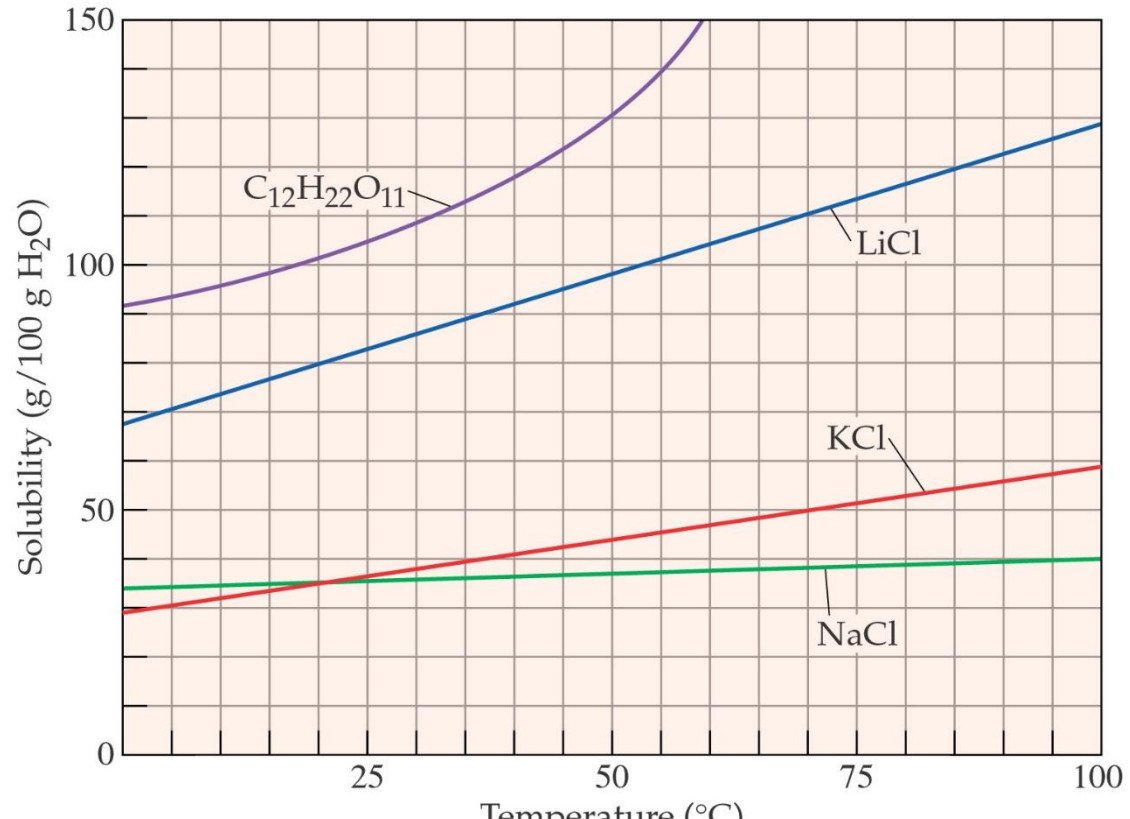
Label the above and below the line blue boxes as unsaturated or supersaturated. Discuss your reasoning as a group, then check your understanding by moving all blue boxes)



Explore: Solubility and temperature

For this graph, answer on the next slide:

- 1) What is the solvent and its quantity?
- 2) What is the solubility of LiCl at 50°C?
- 3) At what temperature is a solution with KCl considered saturated at 50/100g H₂O?
- 4) At what minimum concentration will NaCl be considered supersaturated regardless of temperature?



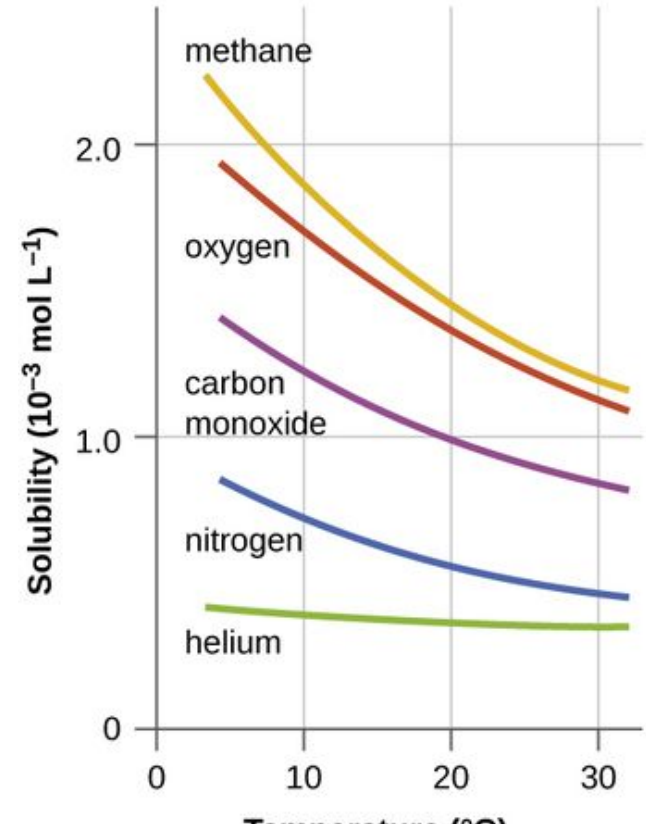
Room for Graph Answers

*** include your explanations**

Explore: Solubility and temperature

The following compounds are all gases.

- What do you notice that is different about the trends of gases as compared to trends of ionic solids?
- Discuss with your group why this might be (feel free to write in your thoughts)

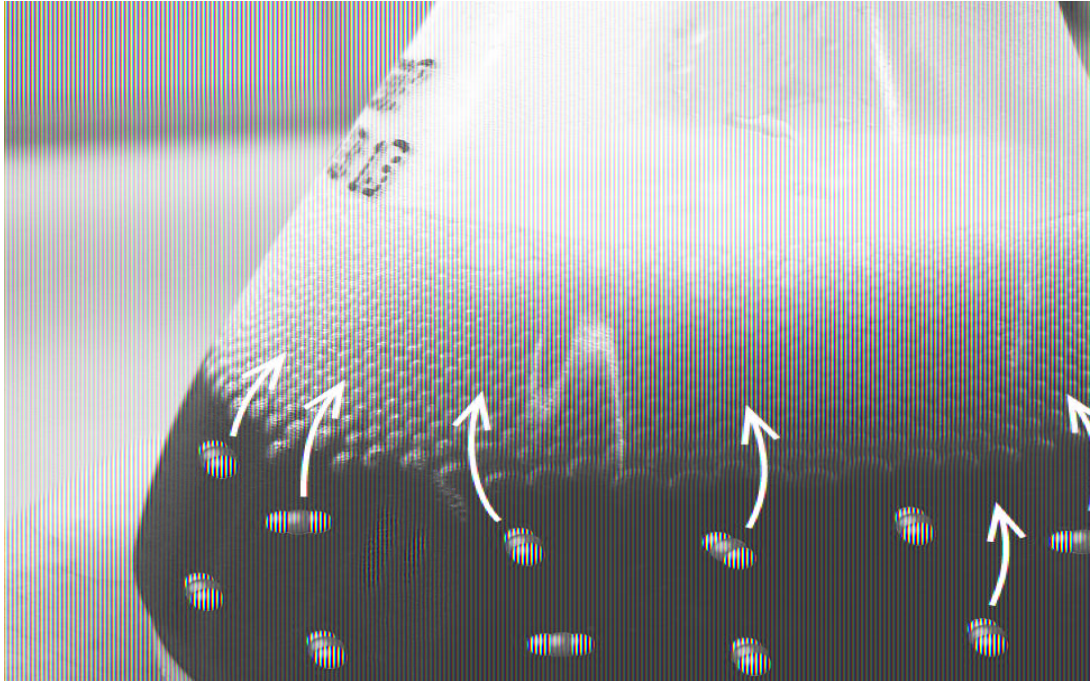


Explore: Solubility and temperature

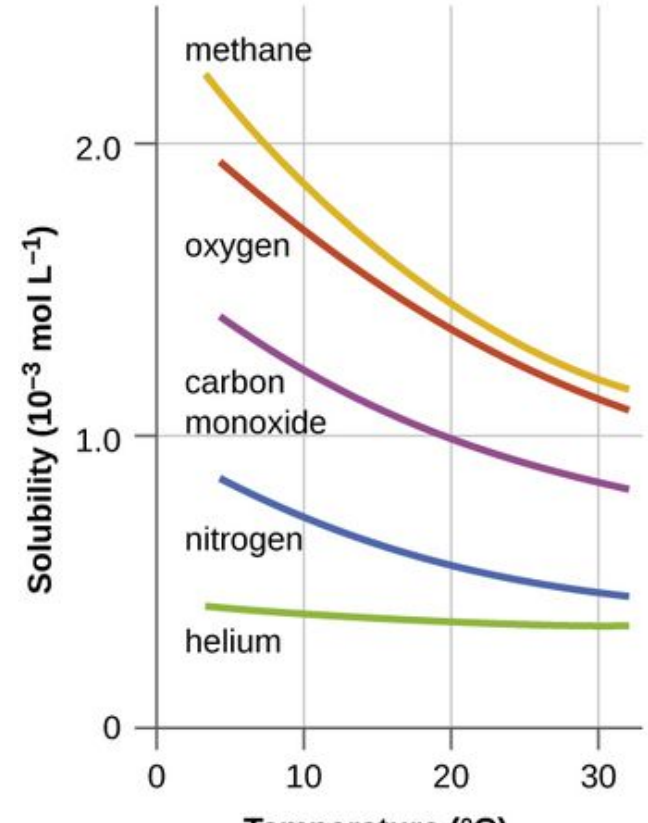
Can you predict what will happen when Grandma drops the mentos into her pop? Can you explain it! (1 minute)



Explore: Solubility and temperature



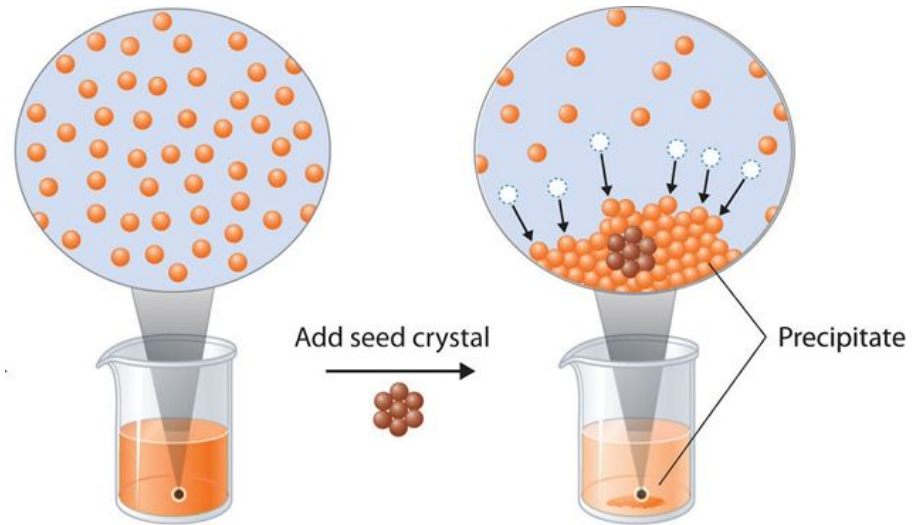
Opening the bottle of carbonated beverage reduces the pressure of the gaseous carbon dioxide above the beverage. The solubility of CO_2 is thus lowered, and some dissolved carbon dioxide may be seen leaving the solution as small gas bubbles.



Supersaturated Solutions

These solutions **contain more dissolved solute than a saturated solution at the same temperature**

- These can be created by heating them up, and allowing them to cool, or using pressure if the solute is a gas
- If done carefully, the dissolved solute will stay dissolved as the solution cools
- These solutions are **unstable** and if disturbed, the excess solute will be removed from solution
 - Example: adding seed crystal or using physical force



Part 3: Apply

Supersaturated Solutions

Recall (or revisit) the video of cold ice from the engage section.

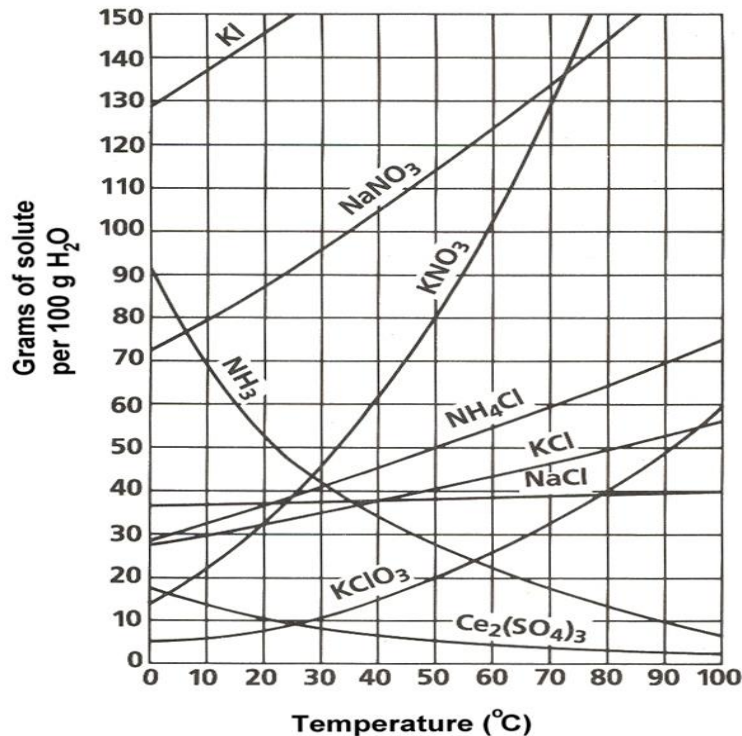
Can you now identify what type of solution was used and what occurred as it was poured?
Explain!

Solubility Curve Questions

PLEASE
CHECK
IN

1. What is the solubility of KCl at 70°C?
2. How many grams of KClO_3 will dissolve in 100 grams of water at 40°C ?
3. How many grams of KClO_3 will dissolve in 50 grams of water at 40°C?

Share your answers with your teacher!
Check answer slides at the end.



Factors Affecting Rate of Dissolving

The factors that affect rate of dissolving are:

- Stirring
- Temperature
- Surface area

Choose one (1) of these factors and at the boards, draw diagram(s) that includes IMF to explain how they affect the rate of dissolving. Take a picture for the next slide

Factor(s) of dissolving Board Photo

Collaborate!

Share your factor drawing with another group.
Record an explanation here of at least one other factor's impact on dissolving after learning from another group.

Factors Affecting Rate of Dissolving

Watch the video on the right.

For your chosen factor, make additional annotations or adjustments in a different colour based on any new information you needed to add to complete your response.



Continue Practice!

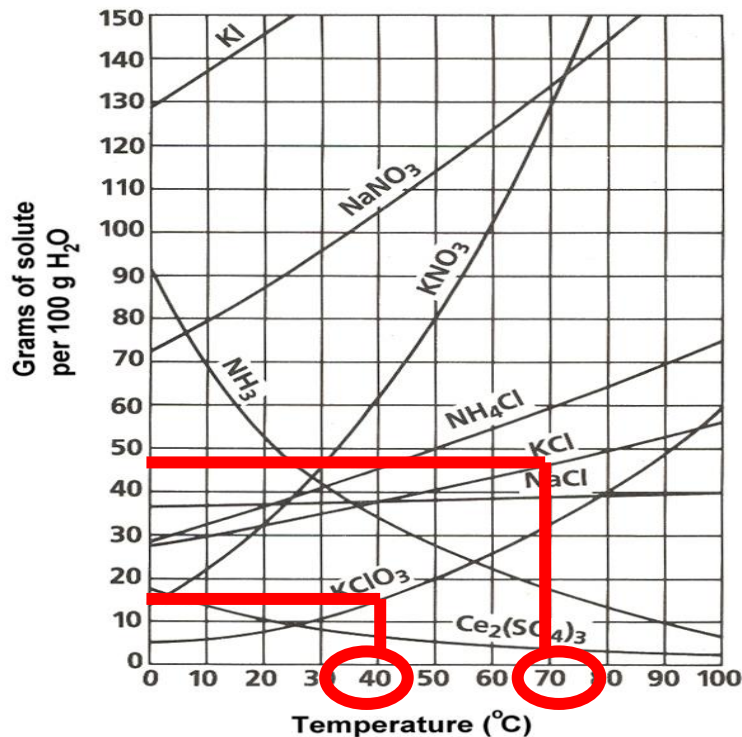
Complete any practice work off the unit plan to solidify your understanding.

Next few slides provide answers to activities

Solubility Curve Questions

1. What is the solubility of KCl at 70°C? **47 g**
2. How many grams of KClO_3 will dissolve in 100 grams of water at 40°C ? **15 g**
3. How many grams of KClO_3 will dissolve in 50 grams of water? **7.5g**

$$\frac{15 \text{ g } \text{KClO}_3}{100 \text{ g H}_2\text{O}} = \frac{x}{50 \text{ g H}_2\text{O}}$$



Part 4: Beyond the course extensions {optional - none avail currently}